

```
ht@ext:~/sambashare/rajni$ source venv/bin/activate
```

```
(venv) ht@ext:~/sambashare/rajni$ python
```

```
Python 3.10.12 (main, Jun 7 2023, 12:45:35) [GCC 9.4.0] on
linux
Type "help", "copyright", "credits" or "license" for more
information.
```

If you want to deactivate the virtual environment, simply write the command in the terminal as:

```
(venv) ht@ext:~/sambashare/rajni/venv/code$ deactivate
```

ReactPy is typically installed using *pip* along with one of the back-ends that it comes natively supported with. You can execute ReactPy as shown in Figure 2 if you want to use Starlette as the back-end.

Let's take a simple example of ReactPy as shown below:

```
1 from reactpy import component, html, run
2 @component
3 def Show ():
4 return html.h1("Hello, world!")
```

```
5 run (Show,host="192.168.11.19",port=7000)
```

You can run this code using the Sublime editor or just the usual Python script. Save the content to *hello.py* and execute it with the following command:

```
$ python hello.py
```

Figure 3 shows how your output should appear.

Enter the web address *http://192.168.11.19:7000* into your browser. As shown in Figure 4, the 'Hello world' message is visible. If an error has occurred, change a small portion of the code, such as:

```
$ run(Show, host="0.0.0.0", port=7000)
```

...or:

```
$ run(Show)
```

If you cannot mention the host and port, then the application is running at the default IP address.

I'll break down what each line does here.

1. The first line, 'from reactpy import component,

```
(venv) ht@ext:~/sambashare/rajni/venv/code$ pip install "reactpy[starlette]"
Requirement already satisfied: reactpy[starlette] in /home/ht/sambashare/rajni/venv/lib/python3.10/site-packages (1.0.1)
Requirement already satisfied: anyio>=3 in /home/ht/sambashare/rajni/venv/lib/python3.10/site-packages (from reactpy[starlette]) (3.7.0)
Requirement already satisfied: asgiref>=3 in /home/ht/sambashare/rajni/venv/lib/python3.10/site-packages (from reactpy[starlette]) (3.7.2)
Requirement already satisfied: colorlog>=6 in /home/ht/sambashare/rajni/venv/lib/python3.10/site-packages (from reactpy[starlette]) (6.7.0)
Requirement already satisfied: fastjsonschema>=2.14.5 in /home/ht/sambashare/rajni/venv/lib/python3.10/site-packages (from reactpy[starlette]) (2.17.1)
Requirement already satisfied: jsonpatch>=1.32 in /home/ht/sambashare/rajni/venv/lib/python3.10/site-packages (from reactpy[starlette]) (1.33)
Requirement already satisfied: lxml>=4 in /home/ht/sambashare/rajni/venv/lib/python3.10/site-packages (from reactpy[starlette]) (4.9.2)
Requirement already satisfied: mypy-extensions>=0.4.3 in /home/ht/sambashare/rajni/venv/lib/python3.10/site-packages (from reactpy[starlette]) (1.0.0)
Requirement already satisfied: requests>=2 in /home/ht/sambashare/rajni/venv/lib/python3.10/site-packages (from reactpy[starlette]) (2.31.0)
Requirement already satisfied: typing-extensions>=3.10 in /home/ht/sambashare/rajni/venv/lib/python3.10/site-packages (from reactpy[starlette]) (4.7.0)
Collecting starlette>=0.19.0 (from reactpy[starlette])
  Downloading starlette-0.20.0-py3-none-any.whl (68 kB)
    +-----+
Collecting uvicorn[standard]>=0.19.0 (from reactpy[starlette])
  Using cached uvicorn-0.22.0-py3-none-any.whl (58 kB)
Requirement already satisfied: idna>=2.8 in /home/ht/sambashare/rajni/venv/lib/python3.10/site-packages (from anyio>=3->reactpy[starlette]) (3.4)
Requirement already satisfied: sniffio>=1.1 in /home/ht/sambashare/rajni/venv/lib/python3.10/site-packages (from anyio>=3->reactpy[starlette]) (1.3.0)
Requirement already satisfied: exceptiongroup in /home/ht/sambashare/rajni/venv/lib/python3.10/site-packages (from anyio>=3->reactpy[starlette]) (1.1.1)
Requirement already satisfied: jsonpointer>=1.0 in /home/ht/sambashare/rajni/venv/lib/python3.10/site-packages (from jsonpatch>=1.32->reactpy[starlette]) (2.4)
Requirement already satisfied: charset-normalizer<4, >=2 in /home/ht/sambashare/rajni/venv/lib/python3.10/site-packages (from requests>=2->reactpy[starlette]) (3.1.0)
Requirement already satisfied: urllib3<3, >=1.21.1 in /home/ht/sambashare/rajni/venv/lib/python3.10/site-packages (from requests>=2->reactpy[starlette]) (2.0.3)
Requirement already satisfied: certifi<=2017.4.17 in /home/ht/sambashare/rajni/venv/lib/python3.10/site-packages (from requests>=2->reactpy[starlette]) (2023.5.7)
Collecting click>=7.0 (from uvicorn[standard]>=0.19.0->reactpy[starlette])
  Using cached click-8.1.3-py3-none-any.whl (106 kB)
    +-----+
Installing collected packages: click, uvicorn, starlette
Successfully installed click-8.1.3 starlette-0.20.0 uvicorn-0.22.0
```

Figure 2: Installation process

```
(venv) ht@ext:~/sambashare/rajni/venv/code$ python hello.py
2023-07-06T12:29:44+0530 | WARNING | The run() function is only intended for testing during development! To run in production, consider selecting a supported backend and importing its associated configure() function from 'reactpy.backend.<package>' where '<package>' is one of ['starlette', 'fastapi', 'sanic', 'tornado', 'flask']. For details refer to the docs on how to run each package.
2023-07-06T12:29:44+0530 | INFO | Running with starlette.applications.Starlette at http://192.168.11.19:7000
INFO: Started server process [4541]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://192.168.11.19:7000 (Press CTRL+C to quit)
INFO: 192.168.11.33:50655 - "GET / HTTP/1.1" 200 OK
INFO: 192.168.11.33:50655 - "GET /index.alif87a11.js HTTP/1.1" 304 Not Modified
INFO: ('192.168.11.33', 50659) - "WebSocket /_reactpy/stream" [accepted]
INFO: connection open
```

Figure 3: Output of *hello.py*

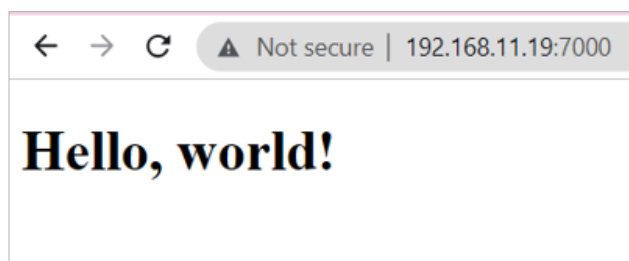


Figure 4: Output of 'Hello, world!' on the browser